

CS 6440 Project Final Submission

Project/Application Name: SkinWise – Final Submission

Term: Spring 2024

Team Name: InnoHealth / Members: Jun Huang, Zhiqiu Jiang, Zifeng Zhang

Section I: Project Overview

Project Title: *SkinWise*

Team Members and Roles/Responsibilities (If Applicable)

- ☐ Team Member 1 Jun Huang – Backend, Deployment
- ☐ Team Member 2 Zhiqiu Jiang – Frontend, User Experience
- ☐ Team Member 3 Zifeng Zhang – Machine Learning, User Interface

TA Mentor: *Matthew Kalita*

External Mentor (If Applicable):

Project Task Status

(Insert your completed Task List as per the Canvas write up here.)

Completed Tasks
Work to form our team.
Team formation and project kick off. Submit Practicum sprint # 1.
Brainstorm team project topic - SkinWise. Assign roles for each team member. Start to explore literature and resources about our project. TA mentor will be assigned.
Project planning and design. Identify the focus of SkinWise as a web-based diagnostic tool. Discuss project topic with our TA mentor. Explore architecture design. Submit Practicum sprint # 2.
Refine project plan. Discuss more details with mentor. Start data processing from the Skin Cancer ISIC Dataset. Set up project environment.
Develop image classification model using deep neural networks. Start the codebase for our web app. Start to implement CRUD Interface for Patient Data Management.
Explore the patient resources from FHIR or available FHIR servers. Refine web app codebase. Mentor check-in # 1. Submit Practicum sprint # 3.

Continue project development. Connect the FHIR skin cancer patient data with the classification model. Web app development.
Testing and deployment of the web app. Mentor check-in # 2. Submit Practicum sprint # 4.
Deployment of the SkinWise web application with CRUD functionality
Mentor check-in # 3. Project finalization phase. Submit Practicum sprint # 5.
Submit Practicum sprint # 6. Submit any other documents/deliverables if needed.

Section II – Project Artifacts

Deployed Application URL: <http://ec2-13-60-53-97.eu-north-1.compute.amazonaws.com:8501>

(A link to your deployed project, Tableau dashboards, or anything else that exists remotely and acts as the user interface for your project.)

GitHub Repository Link: <https://github.gatech.edu/jhuang709/SkinWise/>

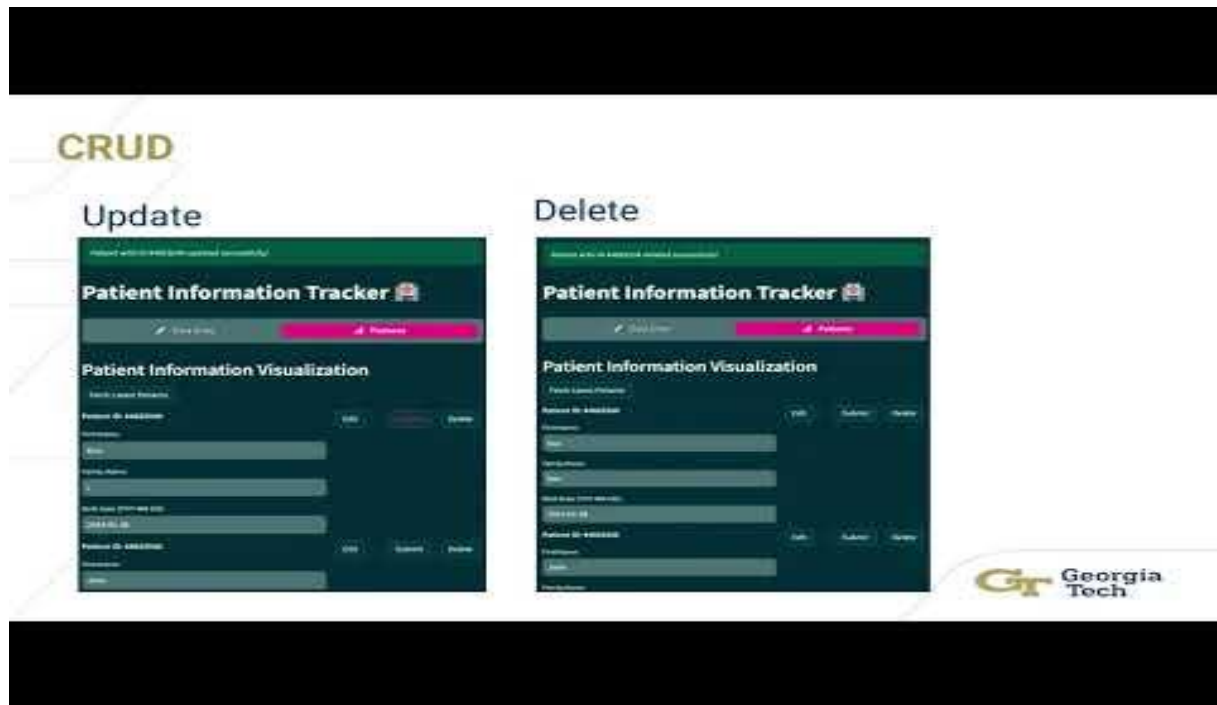
NOTE: All of your code and other project artifacts must be in this repository. This must be a Georgia Tech Enterprise Repository with TA mentors given full access.

GitHub Branch or Commit to Grade (If Applicable): *(If you have a specific commit or branch you would like us to grade instead of main/master, please state it here.)*

Section III – Project Presentation

Please see the Canvas Assignment page for the current semester for general guidelines on the presentation.

Presentation Link: <https://www.youtube.com/watch?v=dDfNKfDLWtM>



Note: Convert **Slides to PDF** and Merge with this single submission. You could add the contents of this deliverable to the final slide deck and convert the whole thing to a PDF to submit. Just make sure all links are clickable and do not include the submission information in the video presentation. (This is to provide a backup to presentations, both to ensure smoother grading in case of an error in a presentation and to maintain a record in Canvas.)

Section IV – Project Documentation

Create a directory in your GitHub repository called “Documentation”. Within this directory provide the following:

<https://github.gatech.edu/jhuang709/SkinWise/tree/main/Documentation>

□ Technical and/or User Manual

<https://github.gatech.edu/jhuang709/SkinWise/blob/main/Documentation/Technical%20and%20User%20Manual.md>

- Provide information on deployment of the application.
- Provide basic user instructions in order to utilize the application fully and show off all features.
- Special Instructions for Grading (“navigate to this URL...type in the following...click on this...etc.”) for the instructing team to follow in order to successfully deploy and run your application (in the event it is necessary, trust us it happens every semester). Include any special build or launch instructions. Please do not forget login information or anything such as FHIR patient IDs that may be needed.

- **PLEASE USE MARKDOWN SO NO DOWNLOAD/CLONE IS REQUIRED.**

- Research
 - https://github.gatech.edu/jhuang709/SkinWise/blob/main/Documentation/Team_SkinWise_IHI_Research.pdf
 - Any research done during the course of the project. This will vary heavily by project. As always, check with your TA mentor on expectations.
- Architecture Diagram
 - <https://github.gatech.edu/jhuang709/SkinWise/blob/main/Documentation/Architecture%20Diagram.png>
 - Include an up to date version of your architecture diagram from Sprint #2.
- Startup and Configuration Files
 - https://github.gatech.edu/jhuang709/SkinWise/tree/main/Documentation/Startup_and_Configuration_Files
 - Dockerfile, docker-compose.yml, etc..., as applicable

For each item included, provide a labeled direct links. Use full URLs, do not embed the link in a word or phrase.

All of the documentation that you submit with your final project should be treated as an official deliverable – this includes the technical/user manual(s). Documentation that is not well structured or clearly written may result in point loss, as it can impact use and thus grading of the application.

SkinWise

CS6440 Spring 2024

Introduction to Health Informatics

Team 23 InnoHealth:

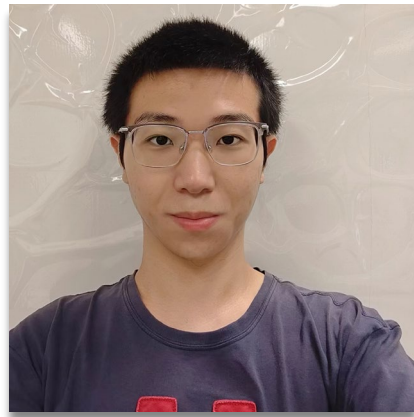
Jun Huang, Zifeng Zhang, Zhiqiu Jiang

Team Member Roles & Contributions



**Jun
Huang**

Project Manager
& Machine Learning
Engineer



**Zifeng
Zhang**

Front-End
Developer & Back-
End Developer



**Zhiqiu,
Jiang**

UX/UI Designer;
Data Scientist

Project Tasks / Activities	Group Member
Ensure smooth functionality of CRUD interface operations, including Create, Delete, etc.	Jun
Ensure seamless functionality of our chosen image classification model within our web app	Zhiqiu, Zifeng
Test the deployment of our web app (locally and also on AWS)	Jun, Zhiqiu, Zifeng
Update our User Interface	Jun, Zhiqiu
Debug the codebase for our web app	Jun
Ensure smooth local deployment of the CRUD interface for Patient Data Management	Jun, Zhiqiu, Zifeng
Ensure the connection between our web app and the FHIR server	Jun

Contributions

Jun: Project management,
development, database & app
deployment, report

Zifeng: Development, PPT and
presentation

Zhiqiu: Development, database &
app deployment and report

Areas of Interest

Machine Learning /AI for Health

Health Analytics & Visualizations

Healthcare Technology, Tools & Scripts

CONTENT

- Introduction
- Implementation Status
- Research
- Demonstration
- Future Work

Introduction

Skin Cancer

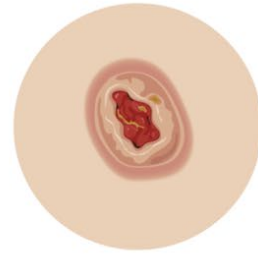
Protect your skin!



Basal cell carcinoma



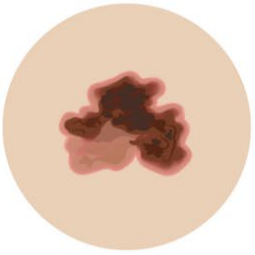
Merkel cell carcinoma



Squamous cell carcinoma



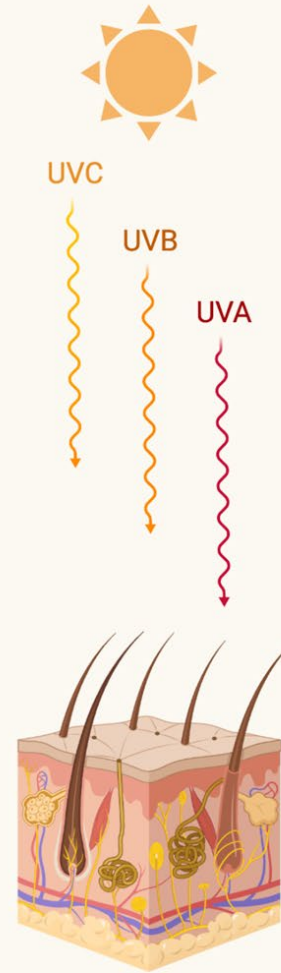
Nodular melanoma



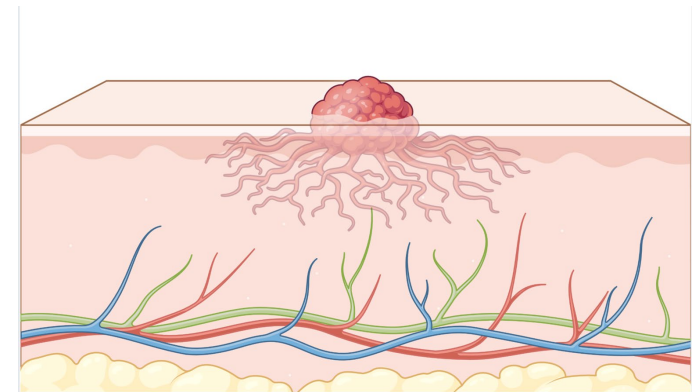
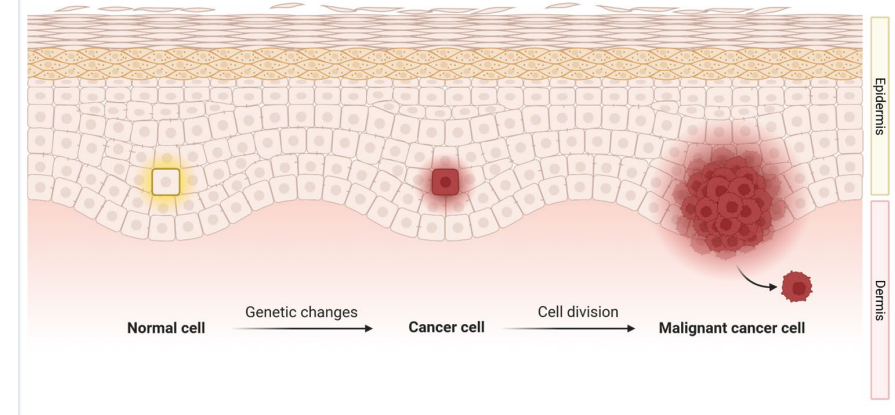
Superficial spreading melanoma



Lentigo maligna melanoma



Skin Cancer Initiation and Progression



Implementation Status

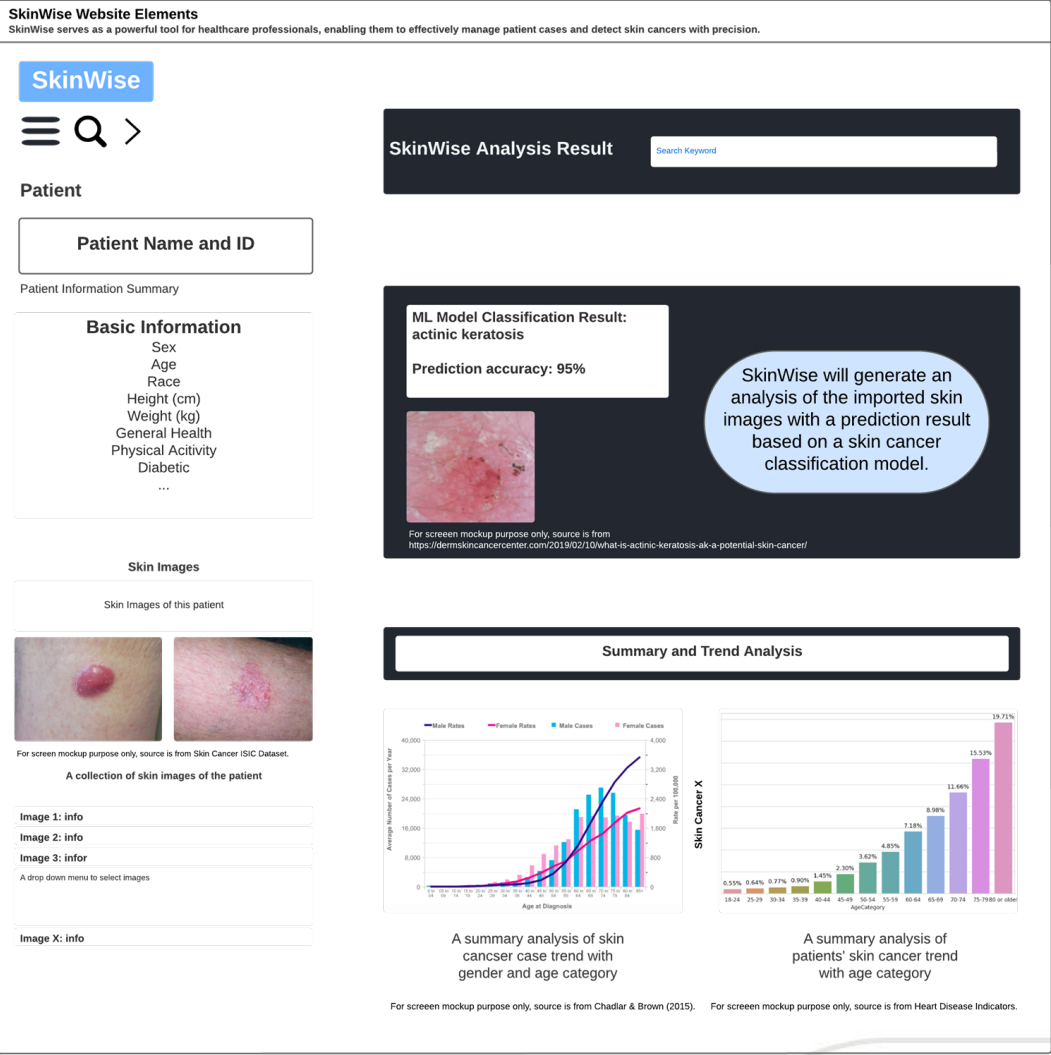
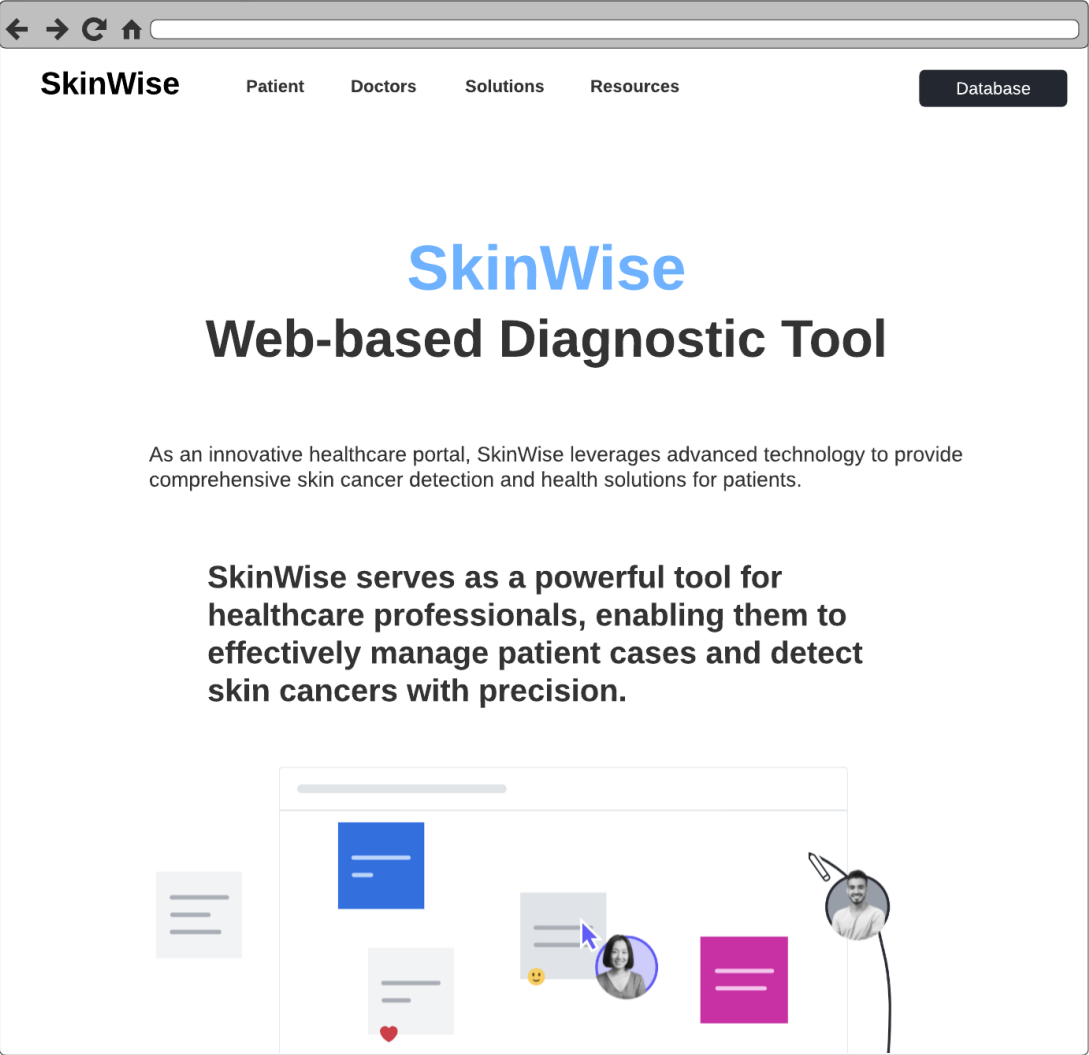
Project Timeline	Project Tasks	Needs	Risks
Jan 8 - Jan 21	Work to form our team.	Identify potential team members with similar project interests and diverse skills and expertise. Establish clear communication channels for team collaboration.	Difficulty in finding suitable team members with the close interests. Communication challenges within the team, e.g., time zones.
Jan 22 - Jan 28	Team formation and project kick off. Submit Practicum sprint # 1.	Formalize team structure and roles. Define project goals and deliverables.	Lack of clarity in project objectives leading to confusion among team members. Initial setbacks in team coordination and workflow.
Jan 29 - Feb 18	Brainstrom team project topic - SkinWise. Assign roles for each team member. Start to explore literature and resources about our project. TA mentor will be assigned.	Research resources and literature relevant to our SkinWise project. Brainstorm the solution. Assign specific roles and responsibilities to team members.	Difficulty in finding relevant and reliable literature/resources. Difficulty in finding relevant health data sources. Potential conflicts arising from role assignments within the team.
Feb 19 - Feb 25	Project planning and design. Identify the focus of SkinWise as a web-based diagnostic tool. Discuss project topic with our TA mentor. Explore architecture design. Submit Practicum sprint # 2.	A project plan outlining the scope, objectives, and timeline of the SkinWise web application. Discussions and consultations with mentor to gather feedback. Explore various architecture design options for the SkinWise. Update our sprint # 2 with mentor's constructive feedback.	Difficulty in finding necessary data sources. Challenges in having interoperability. Shifting project focus leading to scope creep.
Feb 26 - Mar 3	Refine project plan. Discuss more details with mentor. Start data processing from the Skin Cancer ISIC Dataset. Set up project environment.	Detailed project planning and design. Initiate data processing from the Skin Cancer ISIC Dataset. Set up the web app project environment.	Delays in project planning leading to a rushed implementation phase. Technical challenges in processing and handling the dataset. Technical challenges in setting the environment.
Mar 3 - Mar 10	Develop image classification model using deep neural networks. Start the codebase for our web app. Start to implement CRUD Interface for Patient Data Management.	Development of image classification model using deep neural networks. Initial setup of the web app codebase.	Complexity in implementing deep neural networks for image classification. Technical challenges in the web development with connecting with different datasets.
Mar 11 - Mar 17	Call for the patient resources from FHIR. Refine web app codebase. Mentor check-in # 1. Submit Practicum sprint # 3.	Access patient resources from FHIR for integration. Refine web app codebase. Conduct mentor check-in to ensure project alignment.	Delays or issues in obtaining necessary patient resources. Integrating data from FHIR sources can be complex. Relying on FHIR resources introduces dependencies.
Mar 18 - Mar 24	Continue project development. Connect the FHIR skin cancer patient data with the classification model. Web app development.	Continued project development including integration of FHIR data. Progress in web app development.	Difficulties in connecting FHIR patient data with the classification model. Challenges in the development of the web app functionality.
Mar 25 - Mar 31	Testing and validation of the web app. Mentor check-in # 2. Submit Practicum sprint # 4.	Testing and validation of the web app. Final mentor check-in before project submission.	Inadequate testing leading to undiscovered bugs or issues. Insufficient time for addressing feedback from mentor check-in.
Apr 1 - Apr 7	Deployment of the SkinWise web application with CRUD functionality	Deployment of the SkinWise web application using CRUD operations. Examine the deployment online.	Deployment issues causing downtime or functionality failures. Issues with data integrity, such as data loss or corruption, especially during data migration or database updates. Incomplete deployment leading to user dissatisfaction.
Apr 8 - Apr 14	Mentor check-in # 3. Project finalization phase. Submit Practicum sprint # 5.	Finalize project details and address any remaining issues. Prepare for project submission.	Online deployment may result in performance degradation, such as slow response times, increased latency, or system crashes, especially under heavy loads. Other challenges in the deployment such as configuration errors, network issues.
Apr 15 - Apr 21	Submit Practicum sprint # 6. Submit any other documents/deliverables if needed.	Submission of the final project deliverables.	Technical difficulties or errors during submission. Incomplete or insufficient project documentations.

■ Project Tasks

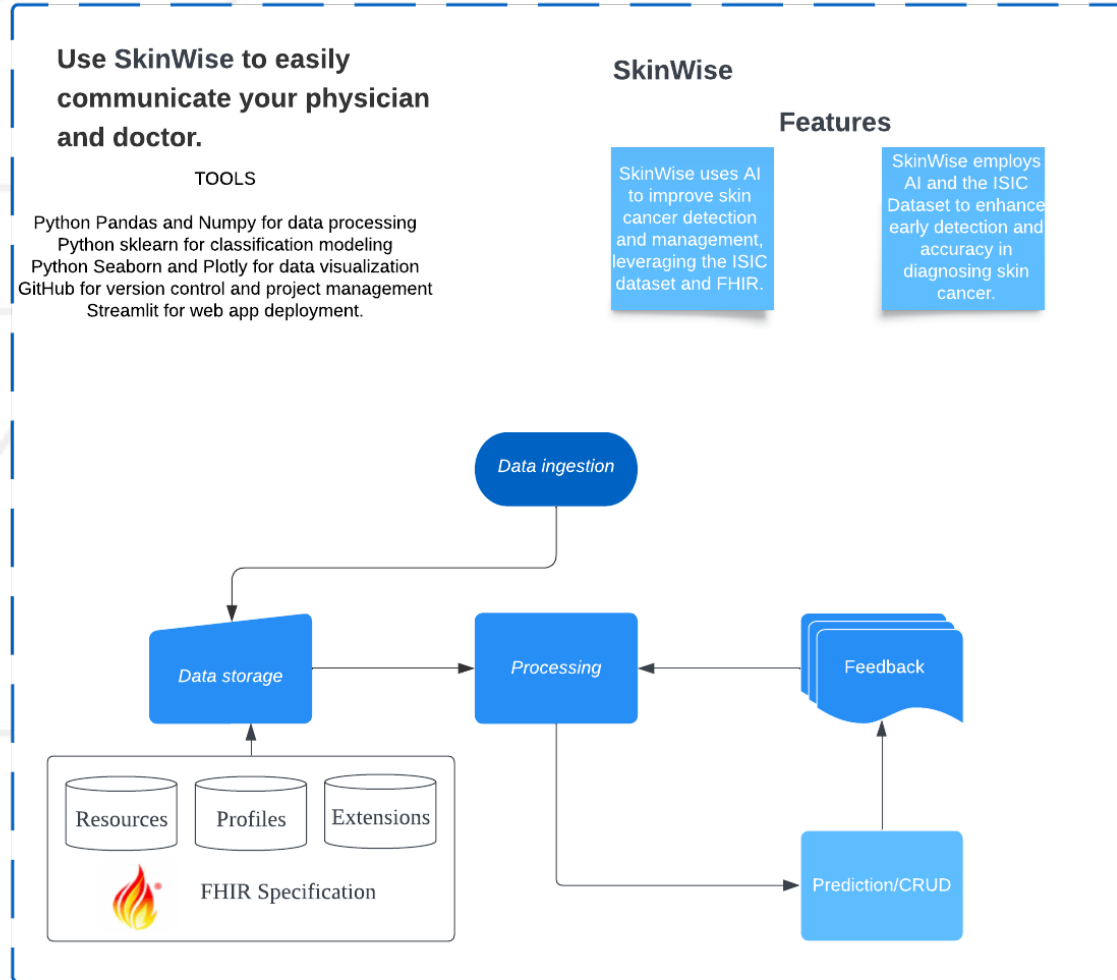
■ Project Timeline

■ Identified Needs And Risks

Mock Up



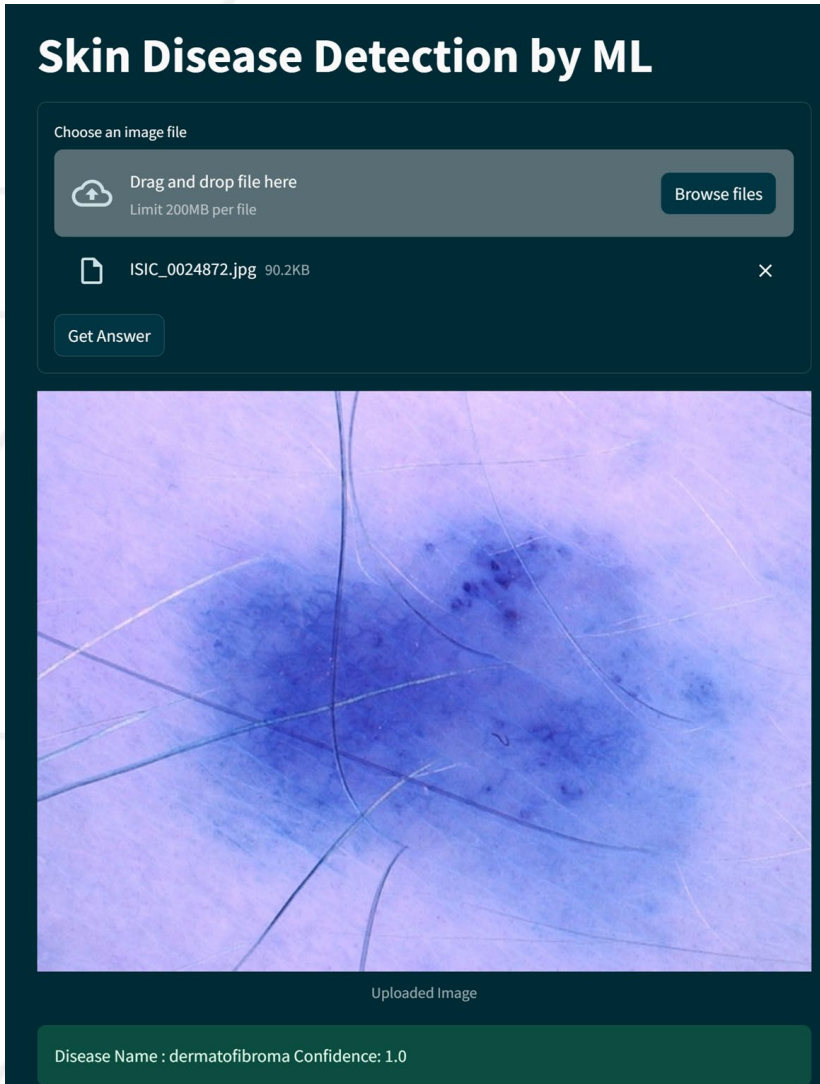
Architecture Diagram



SkinWise Solution Overview Main Components:

- Frontend
- Backend
- Skin Cancer Classification ML Model
- Database
- FHIR Integration for Data Interoperability

Skin disease classification based on Machine learning



Skin disease classification based on Machine Learning (ML).
ML-driven skin disease detection process.

Process Overview:

- User Interaction: Drag and drop image onto the interface.
 - Action: Click to initiate analysis and obtain results.
 - Result Display: Shows the detected disease name and confidence level.
-
- Our SkinWise solution utilizes Machine Learning for skin disease classification.
 - Users can simply drag and drop an image onto the interface for analysis.
 - Upon clicking, the system provides the detected disease name along with the confidence level of the classification.

Local deployment using Docker

The screenshot displays a local development environment with a code editor, a terminal, and a web browser.

Code Editor (VS Code):

- EXPLORER:** Shows the project structure for **SKINWISE**, including files like `__pycache__`, `.idea`, `.streamlit`, `explainer_notebooks` (containing `database.ipynb` and `Plotly_Sankey_Chart.ipynb`), `.gitignore`, `app.py`, `database.py`, `Dockerfile`, `README.md`, `requirements.txt`, `sprint_1_InnoHealth.md`, and PDF files.
- Dockerfile:** The content is as follows:

```
1 # Use the official Python image as a parent image
2 FROM python:3.12-slim
3
4 # Set the working directory in the container
5 WORKDIR /app
6
7 # Copy the dependencies file to the working directory
8 COPY requirements.txt .
9
10 # Install any dependencies
11 RUN pip install --no-cache-dir -r requirements.txt
12
13 # Copy the content of the local src directory to the working directory
14 COPY . .
15
16 # Specify the command to run on container start
17 CMD ["streamlit", "run", "app.py", "--server.port=8501", "--server.address=0.0.0.0"]
18
```
- TERMINAL:** Shows the output of the `docker build` command:

```
=> exporting layers
=> writing image sha256:fdcfb60324e177d70b5d0ea04c93ff2cb03a9b6620a3c904648517
=> naming to docker.io/library/skinwise

What's Next?
View a summary of image vulnerabilities and recommendations → docker scout quick
(base) junhuang@Juns-MacBook-Pro SkinWise % docker run -p 8501:8501 skinwise

Collecting usage statistics. To deactivate, set browser.gatherUsageStats to False.

You can now view your Streamlit app in your browser.

URL: http://0.0.0.0:8501
```

Web Browser (Chrome):

- Address Bar:** Shows the URL `0.0.0.0:8501`.
- Page Title:** **Patient Information Tracker**
- Buttons:** A pink button labeled **Data Entry** is visible.
- Form Section:** Titled **Data Entry for Patient**, it includes:
 - Select Birth Month:** A dropdown menu currently showing **January**.
 - Select Birth Year:** A text input field containing **2024**.
 - Form Fields:** Three input fields labeled **Patient Information**, **Skin Treatment Information**, and **Comment**.

CRUD

Create

Patient Information Tracker

 Data Entry  Patients

Data Entry for Patient

Select Birth Day:
1

Select Birth Month:
January

Select Birth Year:
1925

First Name:
Skin

Family Name:
Wise

Diagnosis:
dermatofibroma

Save Data

Patient data saved to FHIR server successfully!

Read

Patient Information Tracker

 Data Entry  Patients

Patient Information Visualization

Fetch Latest Patients

Patient ID: 44674503

First Name	Family Name	Birth Date (YYYY-MM-DD)
Skin	Wise	1925-01-01
Edit	Submit	Delete

Conditions:
Condition ID: 44674504 - dermatofibroma

CRUD

Update

Patient Information Tracker

 Data Entry  Patients

Patient Information Visualization

Fetch Latest Patients

Patient ID: 44674732

First Name	Family Name	Birth Date (YYYY-MM-DD)
Skin	Wise	1925-01-02

Edit

Submit

Delete

Patient with ID 44674732 updated successfully!

Conditions:

Condition ID: 44674733 - Cancer 

Delete

Patient with ID 44674732 and all associated conditions deleted successfully!

SkinWise: Web-based Diagnostic Tool

SkinWise serves as a powerful tool for healthcare professionals, enabling them to effectively manage patient cases and detect skin cancers with precision.

Patient Information Tracker

 Data Entry  Patients

Patient Information Visualization

Fetch Latest Patients

Demonstration

In this demo video, we showcase the functionalities of our SkinWise web platform.

- Insight into the various features and capabilities
- Key functionalities such as skin disease image analysis, diagnostic suggestions, patient data management, and integration with HAPI FHIR.

Future Work

■ Evolution of SkinWise

- Enhancing the platform's capabilities and extending its utility.

■ Refinement of Deep Learning Model

- Exploring advanced deep learning model to achieve heightened accuracy and sensitivity in skin disease and cancer detection.

■ Expansion of Features:

- Exploring and implementing additional features to bolster comprehensive patient management and follow-up care.

■ User Study:

- Conducting user studies to gather feedback and insights from users.

■ Ongoing Engagement:

- Ongoing engagement with healthcare providers and stakeholders to guide iterative improvements, ensuring alignment with evolving clinical needs and standards.

Thank You!